

AWS re:Invent

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NET305

Network operations, management, and governance best practices

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Questions

- How many of you have some form of network monitoring and logging enabled?
- How many of you spend long cycles to troubleshoot network connectivity issues?
- How many of you are confident in releasing configuration changes to networks?

What will you learn?

- Network monitoring
- Network troubleshooting
- Network management

Network monitoring

Network monitoring consists of four main phases:

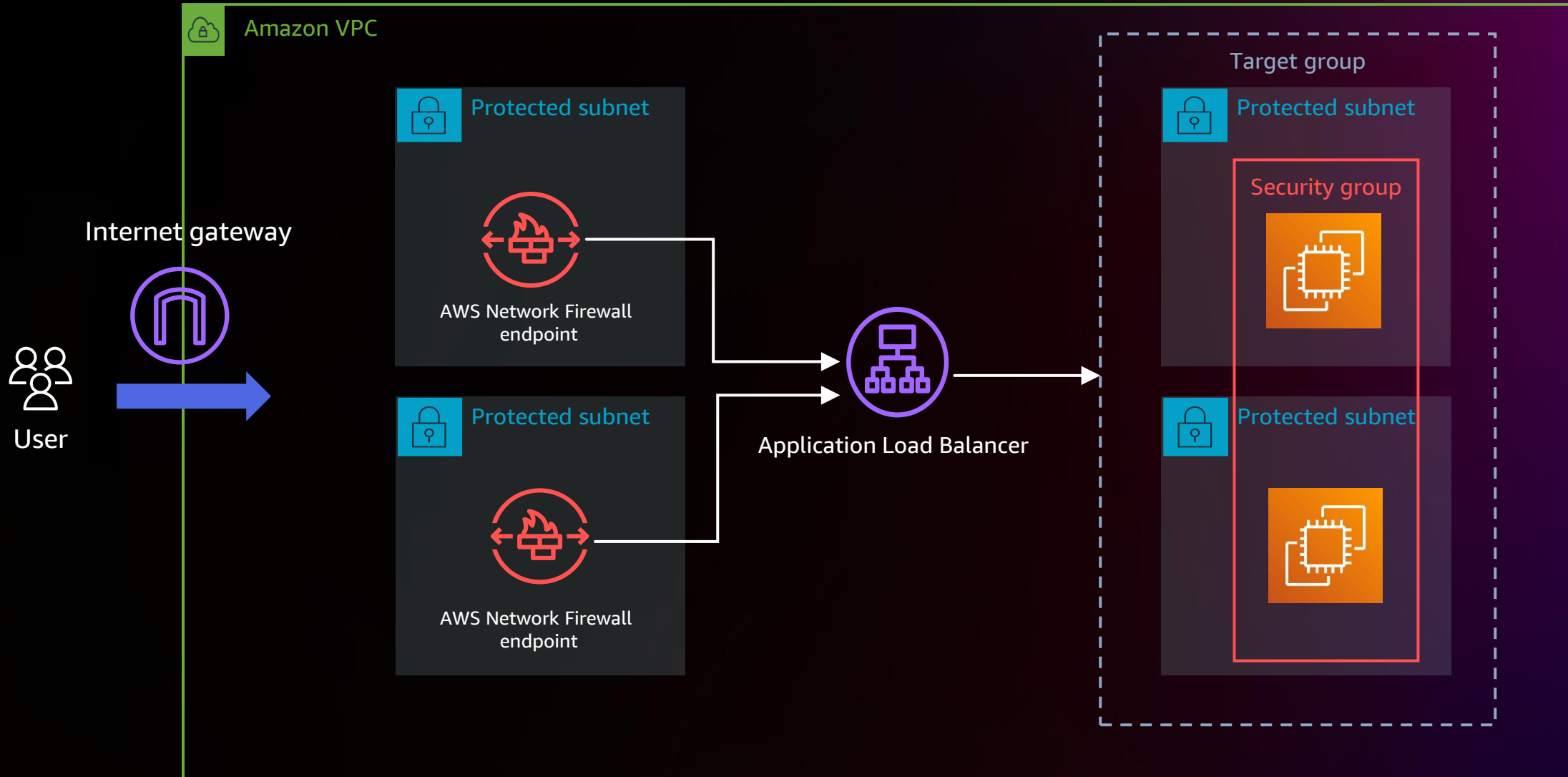
Collect What data should I collect?

Monitor What data should I monitor?

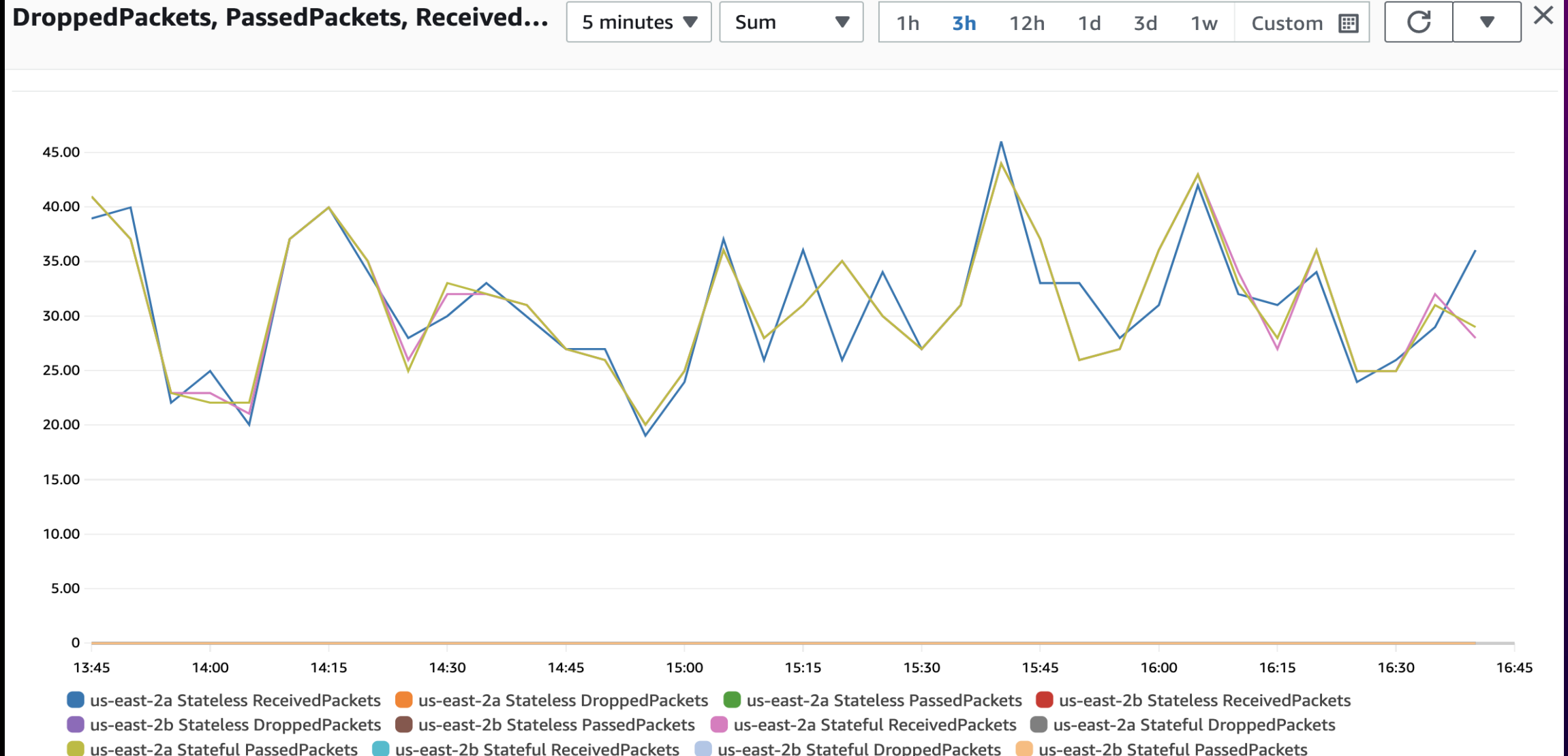
Act How should I act during a network event?
What processes should I follow?

Analyze How should I analyze the monitored data?

Network monitoring example



Network monitoring example

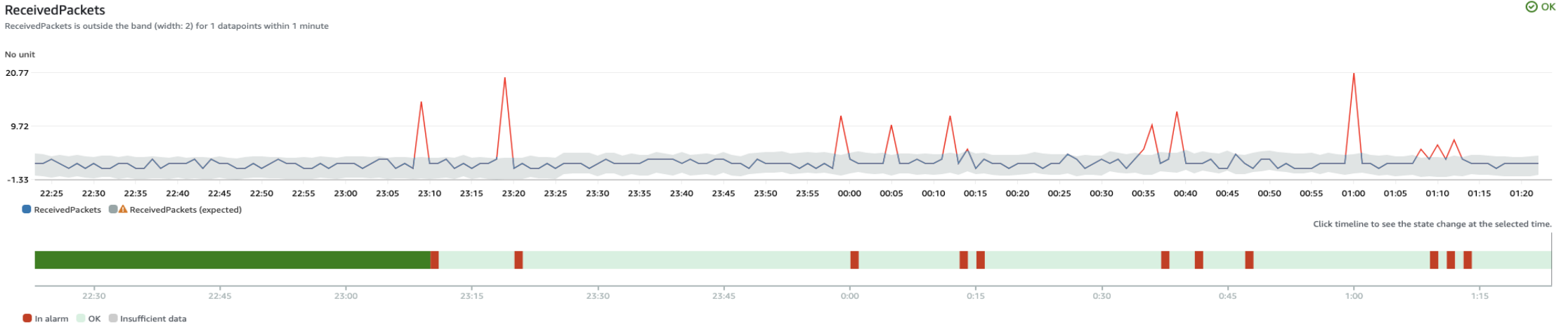


Network monitoring example

```
2022-05-04T16:01:26.000-07:00      {"firewall_name":"network-monitoring-fw","availability_zone":"us-east-2b","event_timestamp":"1651705286...

{
  "firewall_name": "network-monitoring-fw",
  "availability_zone": "us-east-2b",
  "event_timestamp": "1651705286",
  "event": {
    "timestamp": "2022-05-04T23:01:26.169080+0000",
    "flow_id": 1724745131154548,
    "event_type": "netflow",
    "src_ip": "89.248.165.248",
    "src_port": 44267,
    "dest_ip": "10.0.6.28",
    "dest_port": 8443,
    "proto": "TCP",
    "netflow": {
      "pkts": 1,
      "bytes": 40,
      "start": "2022-05-04T22:57:34.908404+0000",
      "end": "2022-05-04T22:57:34.908404+0000",
      "age": 0,
      "min_ttl": 231,
      "max_ttl": 231
    },
    "tcp": {
      "tcp_flags": "02",
      "syn": true
    }
  }
}
```

Network monitoring example



Details Actions History Parent alarms

Details

Name
network-setup-stateful-az-a

Type
Metric alarm

Description
No description

State
OK

Threshold
ReceivedPackets is outside the band (width: 2) for 1 datapoints within 1 minute

Last change
2022-04-24 01:14:06

Actions
Actions enabled

Namespace
AWS/NetworkFirewall

Metric name
ReceivedPackets

FirewallName
network-monitoring-fw

AvailabilityZone
us-east-2a

Engine
Stateful

Statistic
p99

Period
1 minute

Datapoints to alarm
1 out of 1

Missing data treatment
Treat missing data as missing

Percentiles with low samples
evaluate

ARN
arn:aws:cloudwatch:us-east-2:762861022382:alarm:network-setup-stateful-az-a



Network monitoring example

network-monitor-setup

Edit Delete Publish message

Details

Name	network-monitor-setup	Display name	-
ARN	arn:aws:sns:us-east-1:123456789012:network-monitor-setup	Topic owner	123456789012
Type	Standard		

Subscriptions Access policy Delivery retry policy (HTTP/S) Delivery status logging Encryption Tags

Subscriptions (1)

Edit Delete Request confirmation Confirm subscription Create subscription

Search

< 1 > ⚙

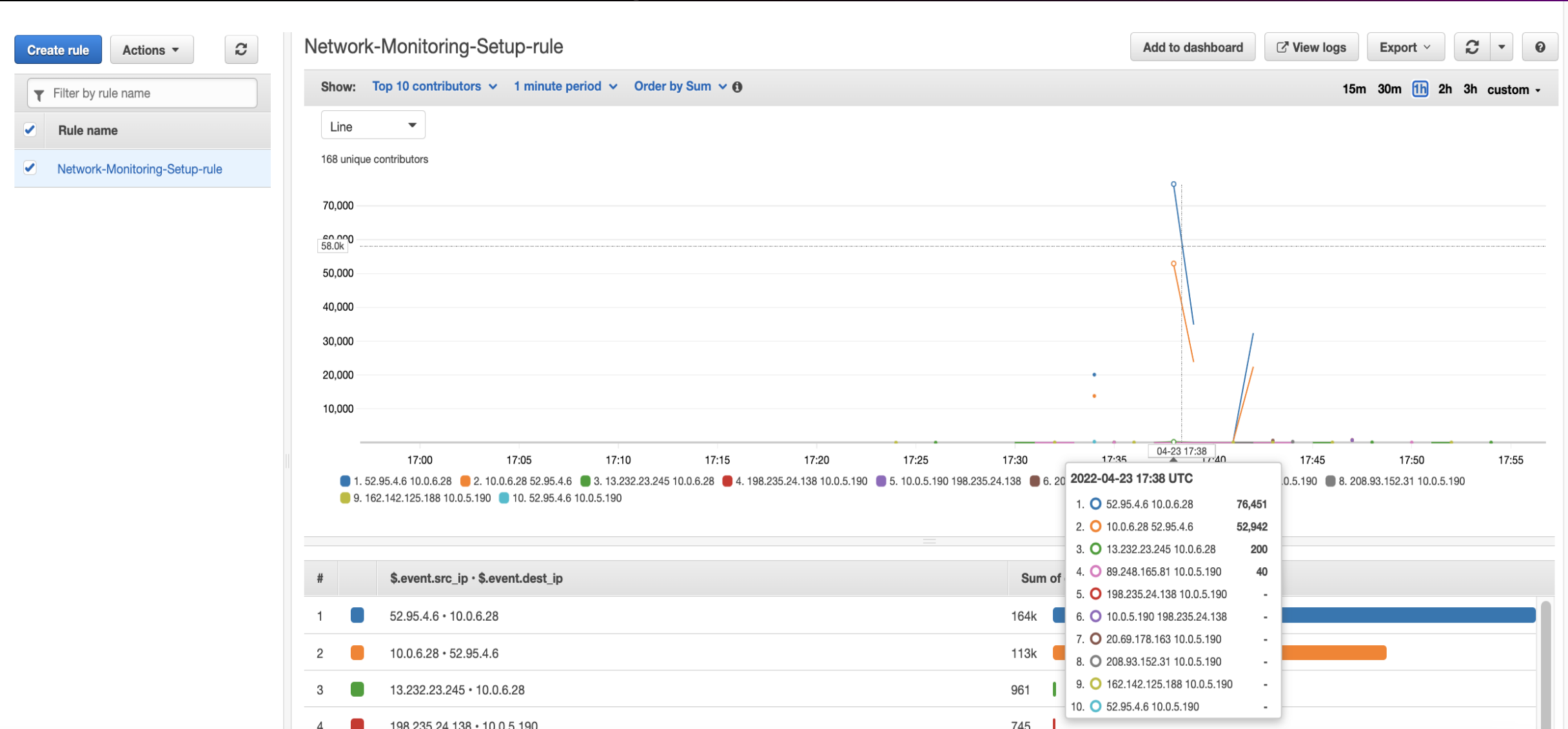
ID	Endpoint	Status	Protocol
d3947d06-2d6a-4905-8168-1bdc961c38b7	123456789012	Confirmed	EMAIL



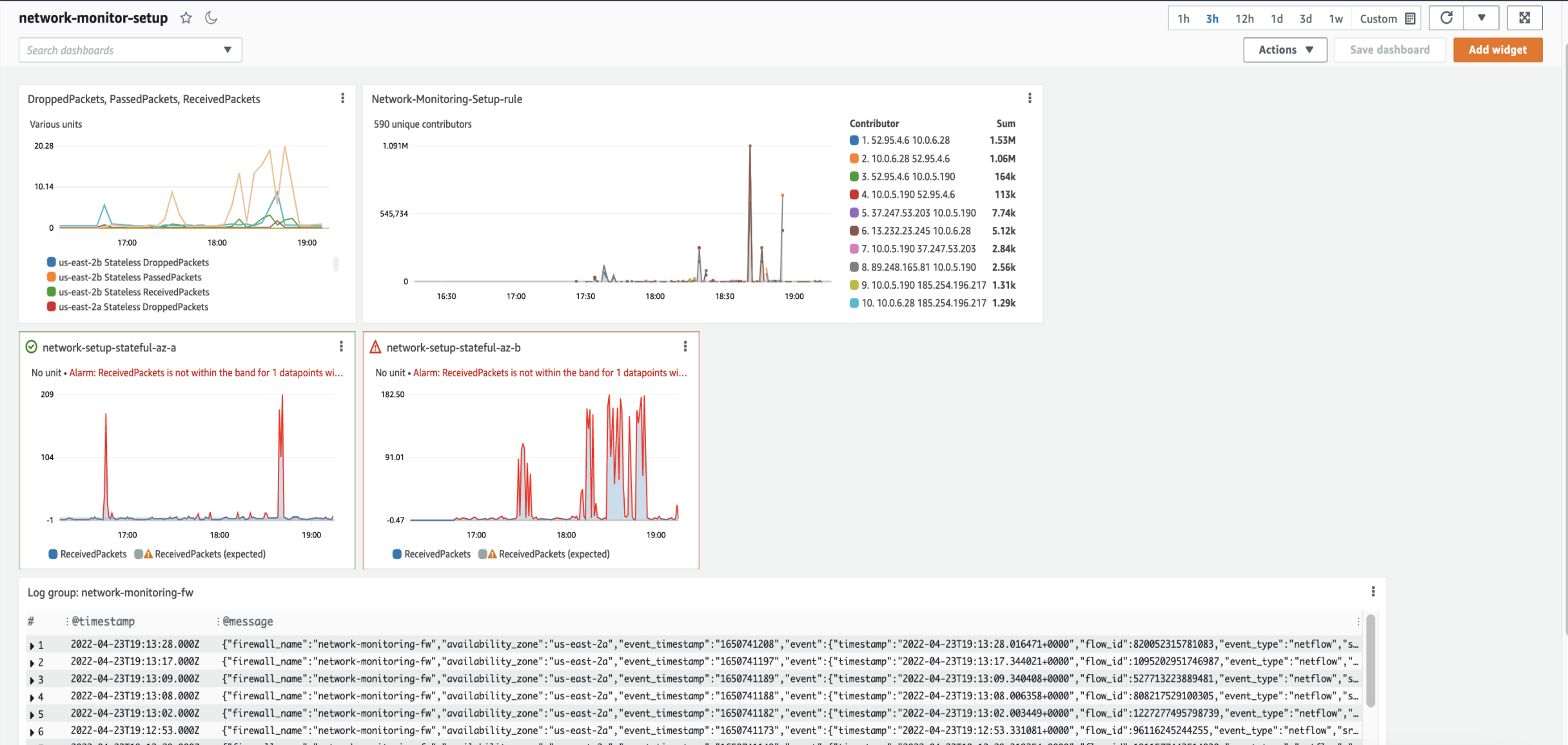
Network monitoring example

Logs Insights

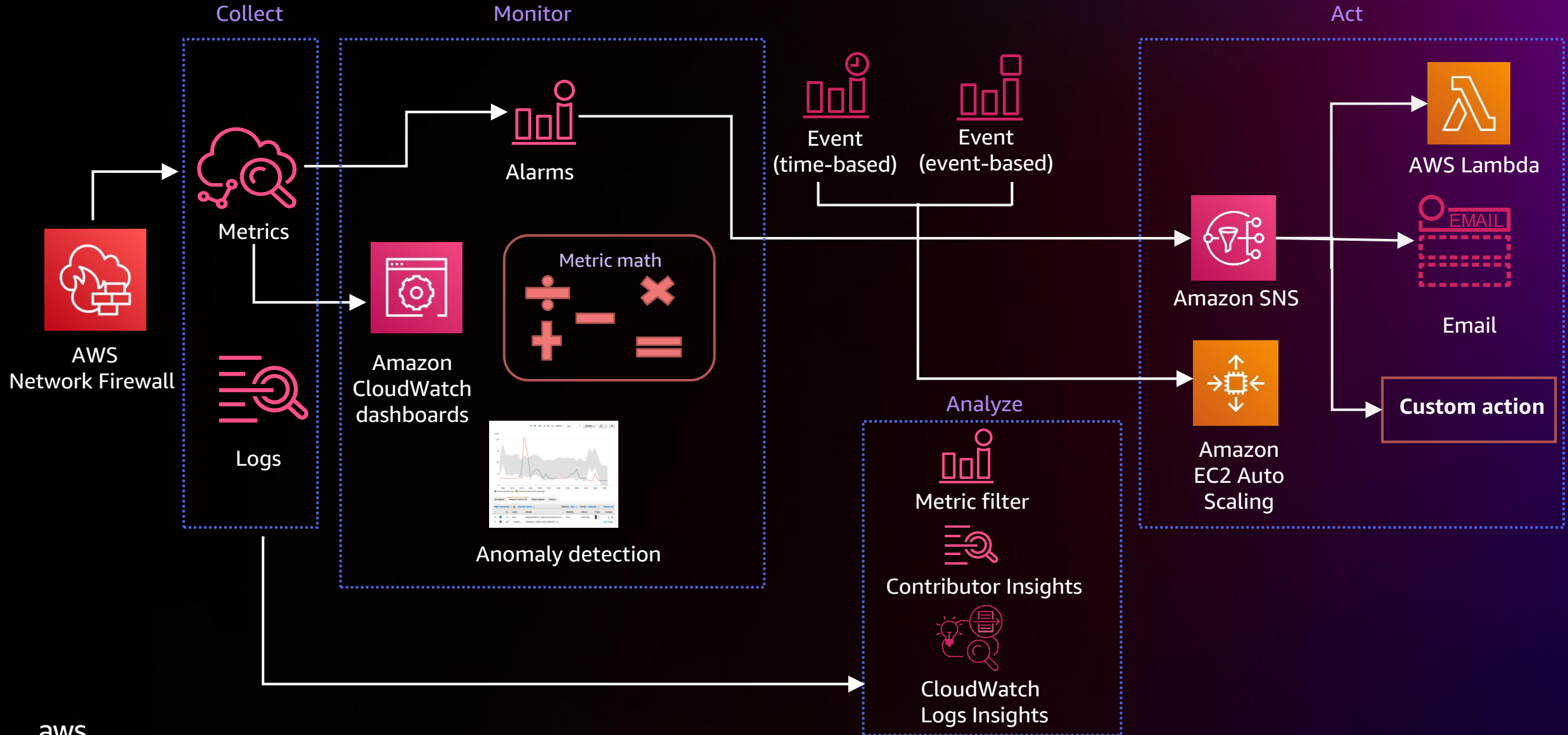
Network monitoring example



Network monitoring example



Network monitoring example



Network monitoring use cases

- Connectivity within AWS
- Connectivity to data centers
- Connectivity to branches
- Connectivity to the internet
- Network security

Questions?



Network troubleshooting

Imagine a Friday evening.

Where do you start? What's the first thing you do?

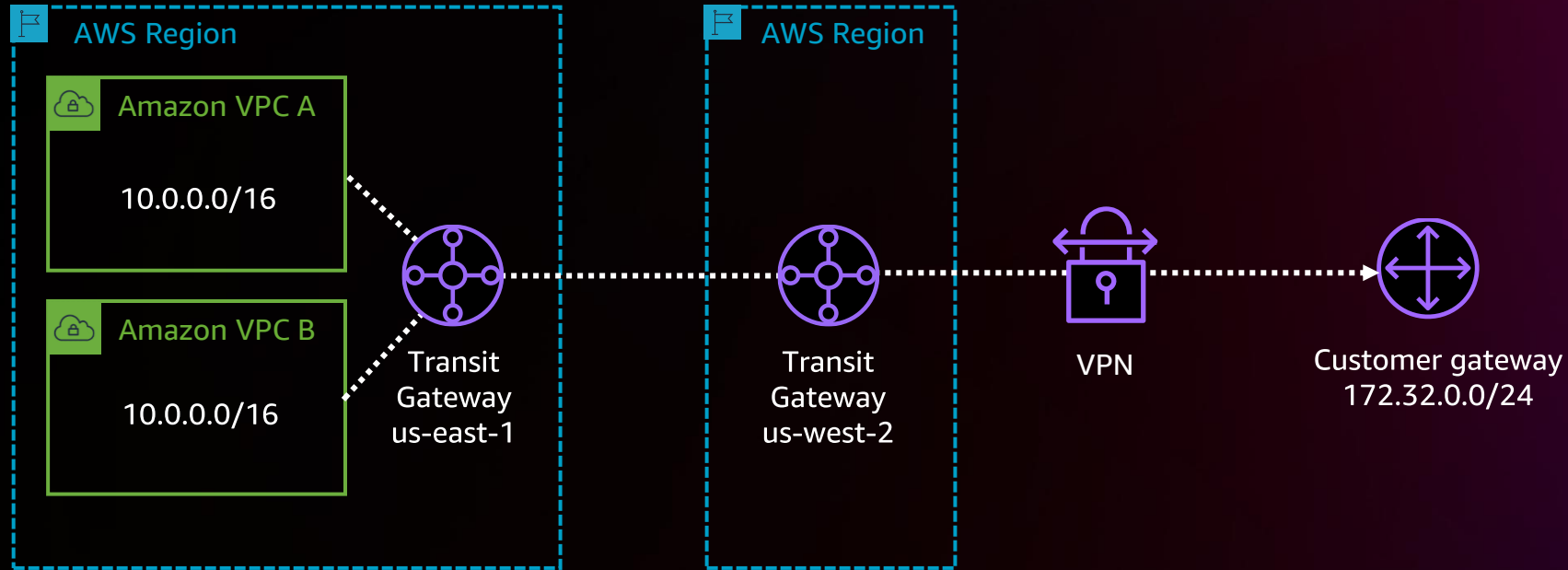
If things stopped working, what's first thing we typically do?

We blame Networking!

Network troubleshooting

- AWS Transit Gateway Route Analyzer
- Amazon VPC Reachability Analyzer
- Amazon Virtual Private Cloud (Amazon VPC) Traffic Mirroring
- Network logging

Transit Gateway Route Analyzer



Transit Gateway Route Analyzer

 Services ▾



Oregon ▾ Support ▾

Forward path

Source	Destination	Status
tgw-attach-092a	tgw-attach-049c	✔ Connected

Return path

Source	Destination	Status
tgw-attach-049c	tgw-attach-092a	⊖ Not connected

 There is no matching route for destination in Transit Gateway route table [tgw-rtb-0f98](#) of Transit Gateway [tgw-011f](#) in region us-west-2.

Source 10.0.0.7

tgw-attach-092a VPC us-east-1

tgw-rtb-00c8 Transit Gateway route table us-east-1 tgw-05e4

tgw-attach-0346 Peering us-west-2

tgw-rtb-0f98 Transit Gateway route table us-west-2 tgw-011f

Destination 172.31.0.8

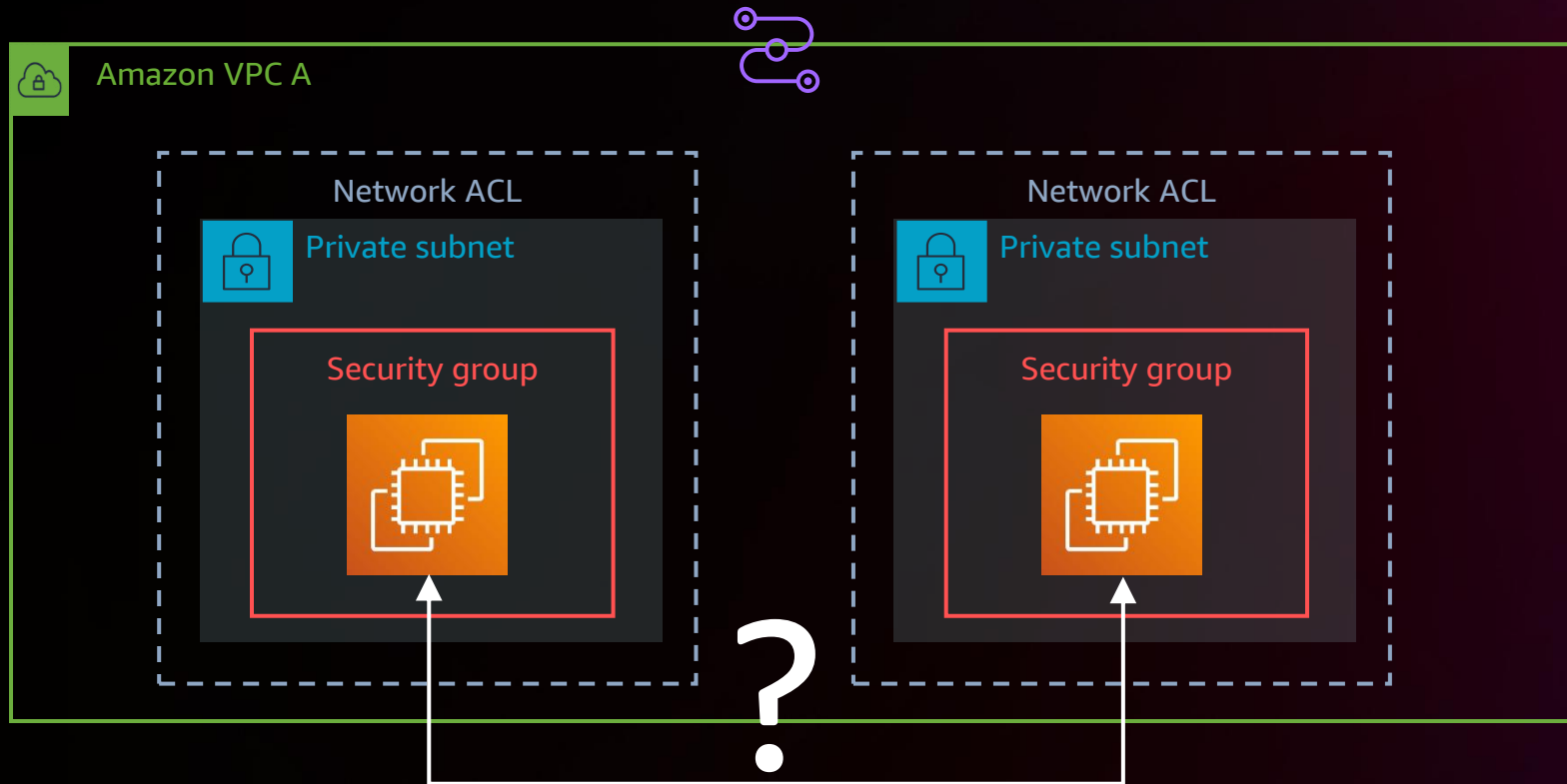
North America VPN (tgw-attach-049c) VPN us-west-2

tgw-rtb-0f98 Transit Gateway route table us-west-2 tgw-011f

Source 10.0.0.7

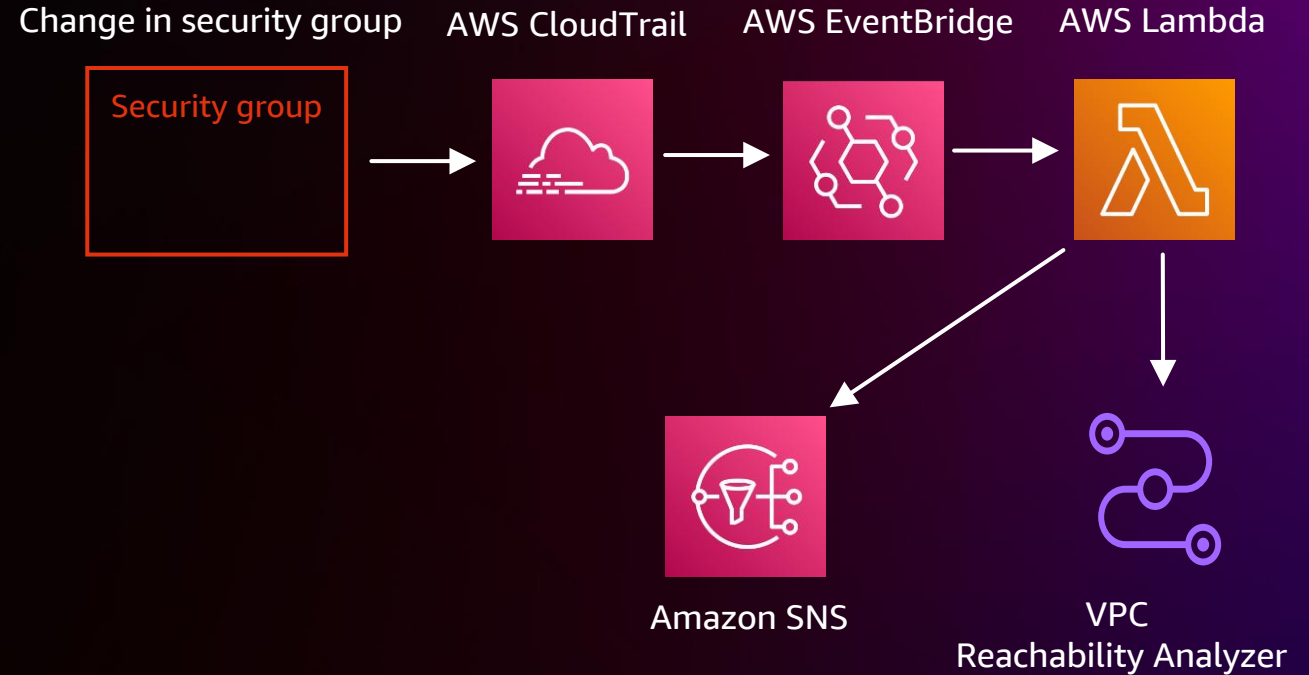
⊖ Not connected

VPC Reachability Analyzer



VPC Reachability Analyzer example

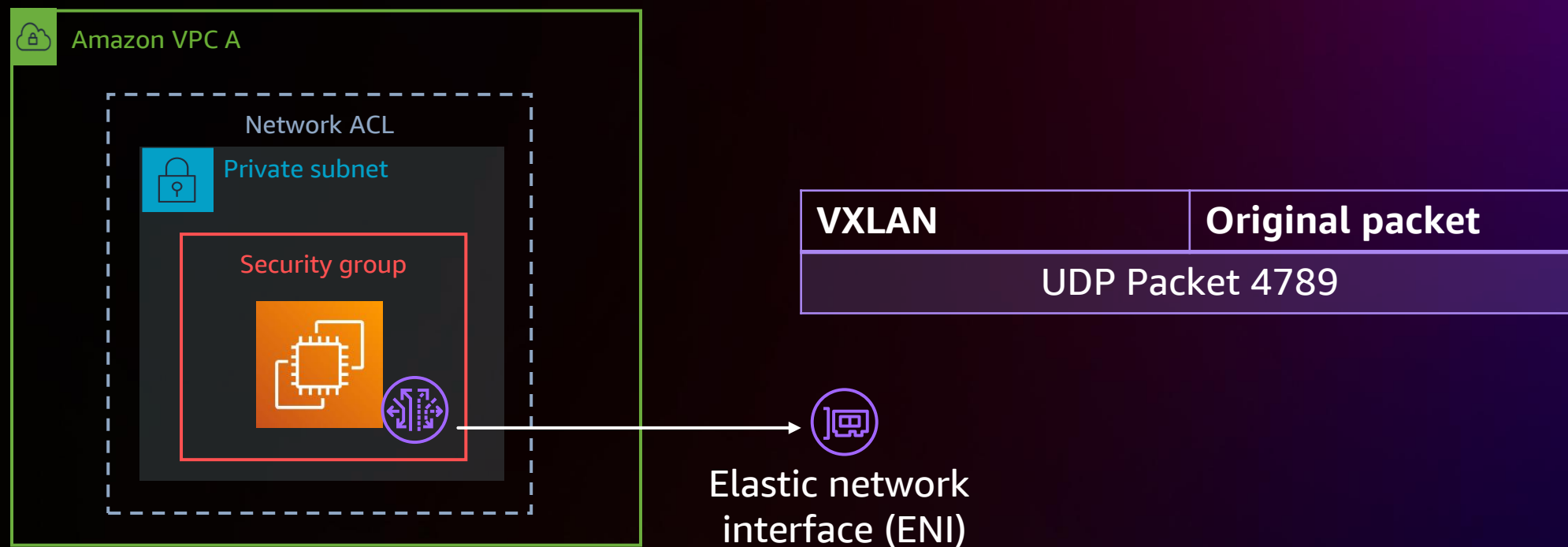
Source		
		<u>i-0f8d73041860f3466</u>
		<u>eni-07c098b2314fb0dcf</u>
		<u>sg-07be75334523a7cdd</u>
		<u>acl-04b3172ce5590ba56</u>
		<u>acl-04b3172ce5590ba56</u>
		<u>sg-07be75334523a7cdd</u>
		<u>eni-02e19b485a9825b16</u>
Destination		
		<u>i-0915bc7f663481da3</u>



VPC Traffic Mirroring

VPC Traffic Mirroring can:

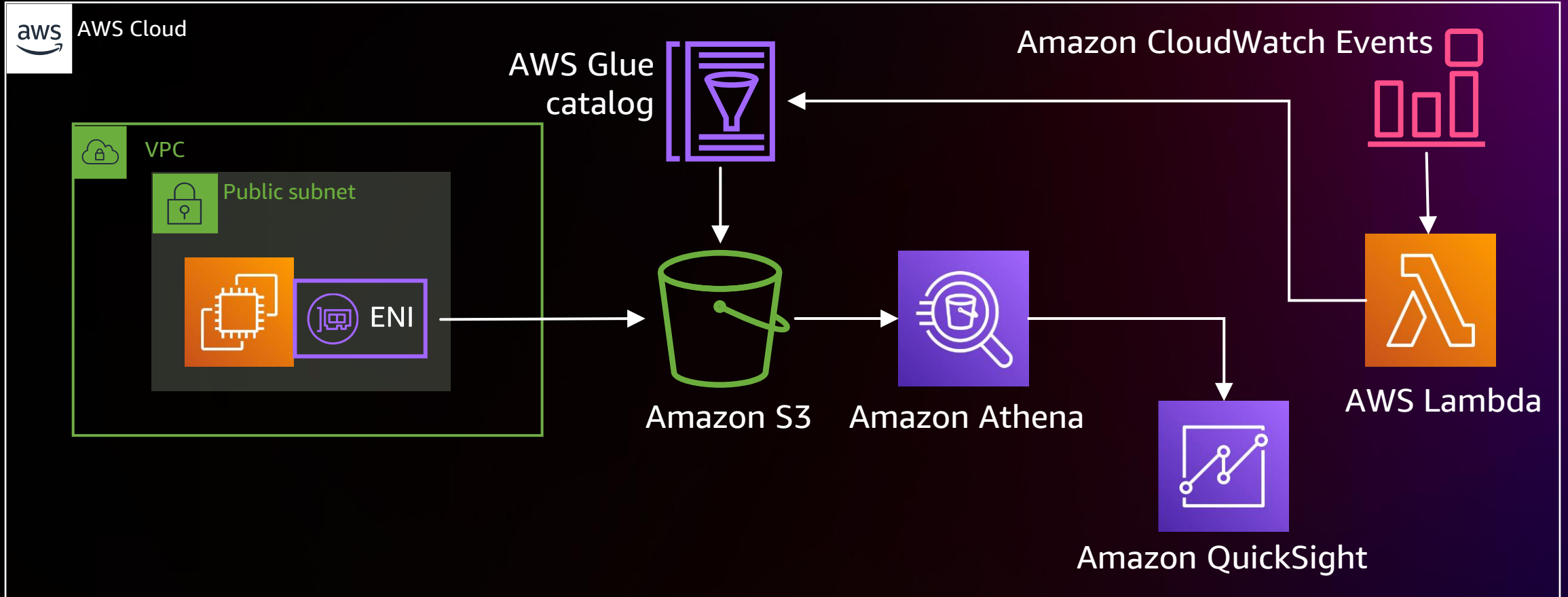
- Copy network traffic from an elastic interface of an EC2 instance
- Send the traffic to out-of-band security and monitoring appliances for content inspection, traffic monitoring, or troubleshooting



Network logs

- VPC Flow Logs
- Transit Gateway Flow Logs
- Site-to-Site VPN connection logs
- Amazon Route 53 Resolver query logs
- AWS Network Firewall logs
- Elastic Load Balancing logs

Troubleshooting VPC Flow Logs example



Summary of network operations

Collect



Metrics



Logs

Monitor



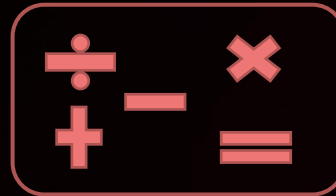
Alarms



Metric filter



CloudWatch
dashboards



Metric math



Anomaly detection

Troubleshoot



VPC Traffic
Mirroring



Flow logs



VPN, ELB, ANF logs



VPC Reachability
Analyzer



Transit Gateway
Route Analyzer



Route 53 Resolver
query logs

Analyze



CloudWatch
Contributor
Insights



CloudWatch Log
Insights



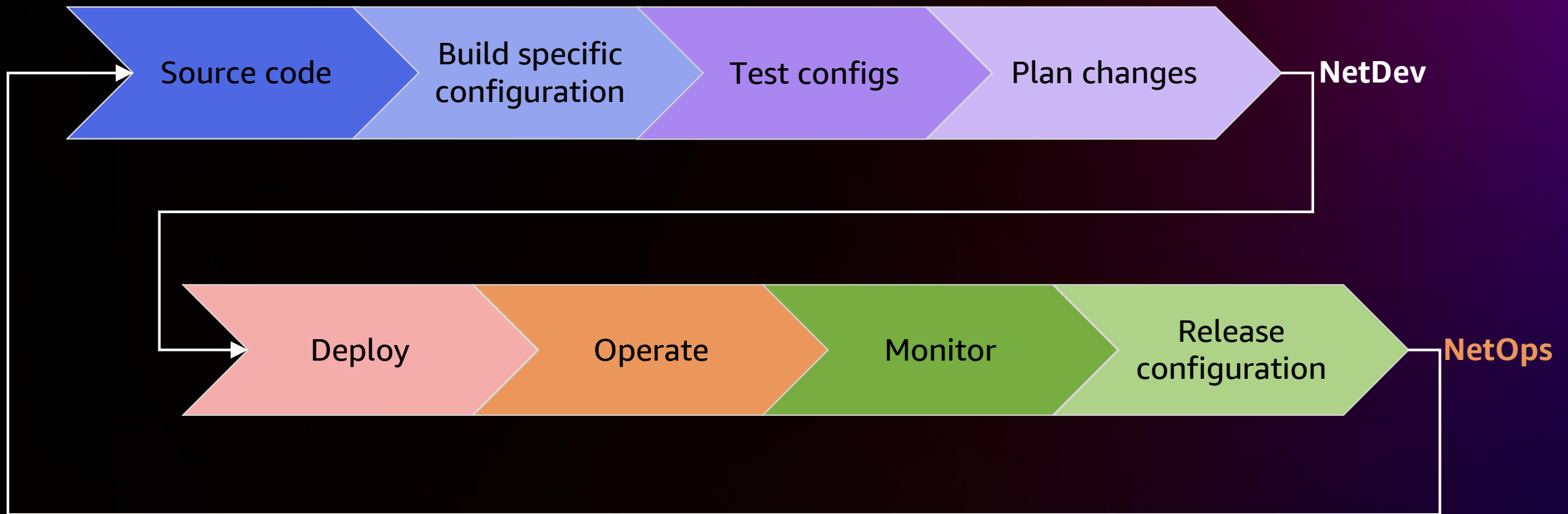
Third-party
solution

Questions?



Network management

NetDevOps  Network development + network operations



AWS CDK

App

Stacks

Resources



```
~/hello-cdk (zsh)
> cdk diff hello-cdk-1
Stack hello-cdk-1
Resources
[-] AWS::SNS::Topic MyFirstTopic MyFirstTopic0ED1F8A4
  [-] DisplayName
    [-] My First Topic Yeah
    [-] Hello, CDK!

~/hello-cdk master* 6s
> cdk deploy hello-cdk-1

hello-cdk-1: deploying...
hello-cdk-1: creating CloudFormation changeset...
0/3 | 12:25:30 PM | UPDATE_IN_PROGRESS | AWS::CDK::Metadata | CDKMetadata
0/3 | 12:25:30 PM | UPDATE_IN_PROGRESS | AWS::SNS::Topic | MyFirstTopic (MyFirstTopic0ED1F8A4)
1/3 | 12:25:31 PM | UPDATE_COMPLETE | AWS::SNS::Topic | MyFirstTopic (MyFirstTopic0ED1F8A4)
2/3 | 12:25:32 PM | UPDATE_COMPLETE | AWS::CDK::Metadata | CDKMetadata
2/3 | 12:25:35 PM | UPDATE_COMPLETE_CLEANUP | AWS::CloudFormation::Stack | hello-cdk-1
3/3 | 12:25:36 PM | UPDATE_COMPLETE | AWS::CloudFormation::Stack | hello-cdk-1

✅ hello-cdk-1

Stack ARN:
arn:aws:cloudformation:us-east-1:585695036304:stack/hello-cdk-1/b9da27f0-fafe-11e9-b7b3-12f2c8f206e2

~/hello-cdk master* 33s
>
```

Core framework

AWS Construct Library

AWS CDK CLI

AWS CDK constructs

L1: Automatically generated

CloudFormation
resources

L2: High-level service constructs

AWS constructs

L3: Opinionated abstractions

Purpose-built
constructs

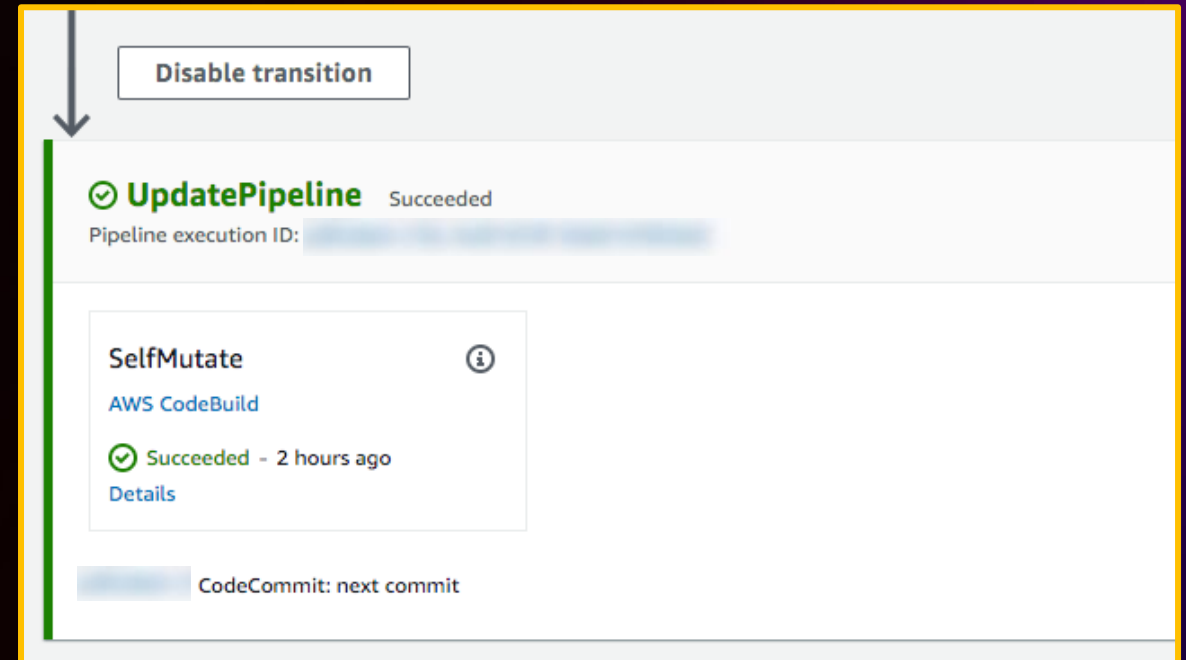
Management of complex network architectures made easy with AWS CDK

Bootstrapping

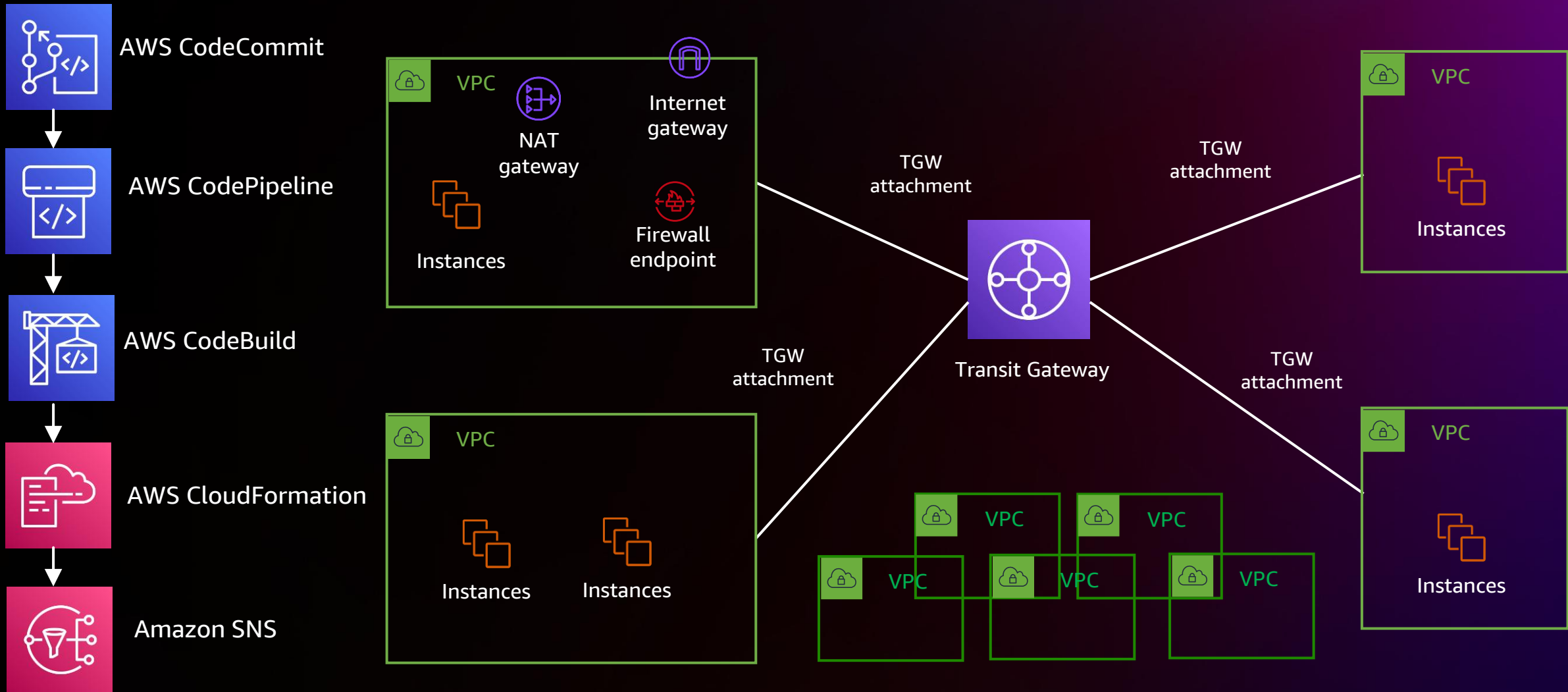
```
export CDK_NEW_BOOTSTRAP=1

cdk bootstrap \
  --profile ADMIN-PROFILE \
  --cloudformation-execution-policies \
    arn:aws:iam::aws:policy/AdministratorAccess \
  --trust PIPELINE-ACCOUNT-ID \
    aws://ACCOUNT-ID/REGION
```

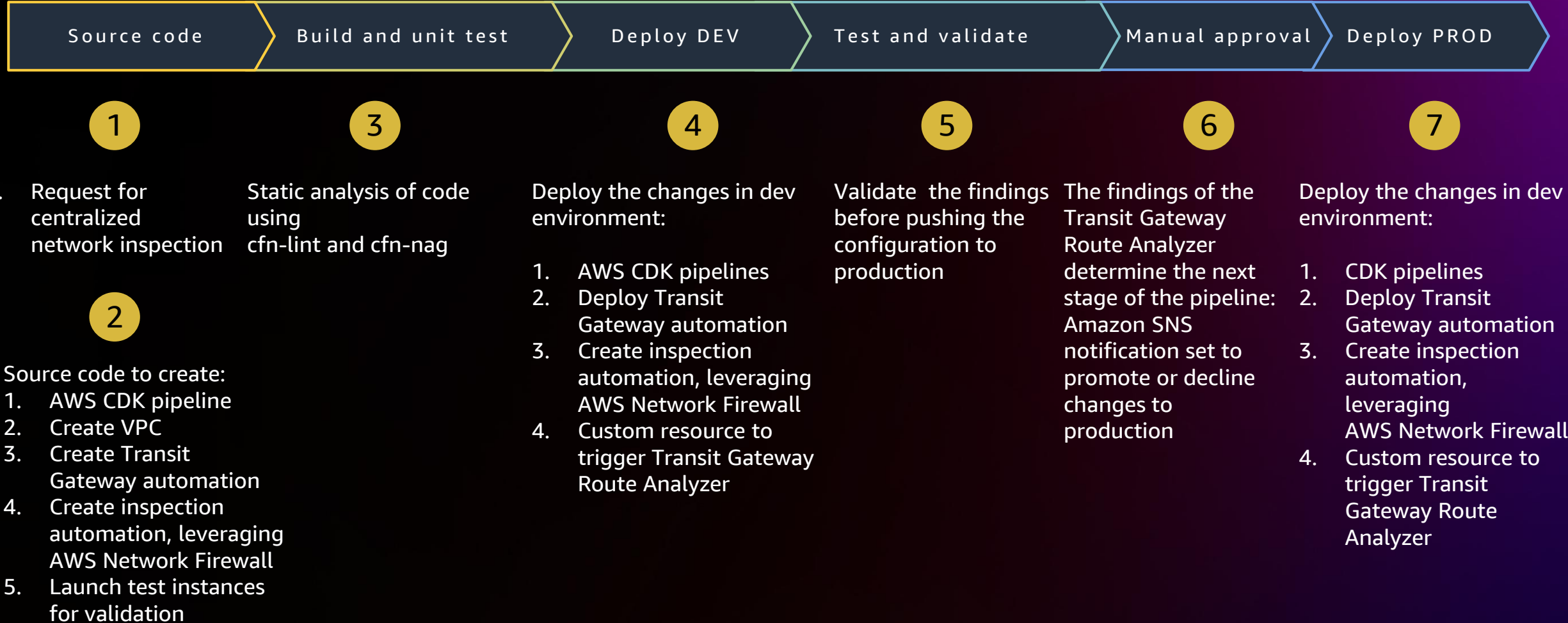
Self-mutating pipeline



Network management example



Network management steps



Key takeaways

Network monitoring

- Collect
- Monitor
- Act
- Analyze

Network troubleshooting

- Transit Gateway Route Analyzer
- VPC Reachability Analyzer
- VPC Traffic Mirroring
- Network logging

Network management

- Pipelines
- Infrastructure as code
- Build
- Test
- Validate
- Deploy

Questions?



Thank you!

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Please complete the session
survey in the **mobile app**



Resources



Introduction to network transformation on AWS: Part 1



Analyse VPC Flow Logs with point-and-click Amazon Athena integration



CDK best practices with AWS CDK



Introduction to network transformation on AWS: Part 2



Analysing VPC Flow Logs using Amazon Athena, and Amazon QuickSight



NetDevOps: A modern approach to AWS network deployments